Day: Thu

Date: 22-Oct-2026

Assg 2 : SQE

Name : Saad Bin Zahid
Reg NO : Sp24-BSE-153
Class : BSE-6A

Questions

Look at the software industry and find out most critical software quality processes. Also find out which of them are automated and using which tools / agents.

The most critical software quality processes are:

Functional Testing
Performance "
Data "
Security "

→ Introduction:

- Software quality processes ensure that software works correctly, performs efficiently, handles data accurately and remains secure.

- In the software industry, four critical software quality processes are Functional Testing, Performance Testing, Data Testing and Security Testing.

- Many of these processes are automated

Date: Day:

using specialized tools to save time, reduce human errors and improve software reliability.

1. Functional Testing

- **Definition:**
  Functional testing checks whether software features work according to the specified requirements.

- **Examples:**
  checking login, registration, search, payment and password reset functionality.

- **Automation:**
  Yes many functional tests can be automated.

- **Tools Used:**
  → Selenium : Automates web browser testing
  → Playwright : Tests web applications across different browsers.
  → Cypress : Automates frontend testing.
  → Postman : Tests APIs and verifies their responses.

Date: Day:

- **Industry Practice:**
  - Developers and QA engineers run automated functional and regression tests whenever software code changes. Manual testing ~~and~~ is also used for exploratory testing and checking complex user experiences.

2. PERFORMANCE TESTING

- **Definition:**
  Performance testing measures the speed, stability, responsiveness and scalability of software under different workloads.

- **Example:**
  Checking whether an online shopping website remains responsive when thousands of users access it simultaneously.

- **Automation:**
  Yes, load, stress and endurance tests can largely be automated.

- **Tools Used:**
  → Apache JMeter: Simulates multiple users and measures application performance.

Date: Day:

→ k6 : Runs scripted performance tests and integrates with CI/CD pipelines.

→ Gatling : Tests application performance under simulated workloads.

→ Grafana : Displays performance metrics through dashboards when connected to suitable data sources.

• **Industry Practice :**

Companies run performance tests before major releases. and use monitoring tools to identify slow responses, high resource usage and system failures.

3- DATA TESTING

• **Definition :**

Data testing verifies that data is accurate, complete, consistent and stored or transferred correctly between systems.

• **Example :**

Checking whether customer records are saved correctly, database calculations are accurate and data remains consistent after migration.

Date: Day:

- **Automation:**

Yes, many data quality checks and database tests can be automated.

- **Tools Used:**

→ SQL: checks database records, missing values, duplicates and incorrect calculations.

→ dbt: Automates data transformation tests.

→ Great Expectations: Defines and validates data quality rules.

→ Soda Core: Helps detect data quality problems through automated checks.

- **Industry Practice:**

Organizations schedule database validation, test data pipelines, and run data quality checks whenever data is imported, transformed or updated.

4. SECURITY TESTING

- **Definition:**

Security testing identifies

Date: Day:

weaknesses that could allow unauthorized access, data theft or attacks on software systems.

- **Example:**
  Checking for SQL injection, weak authentication, unauthorized access and exposed sensitive information.

- **Automation:**
  Yes, many security checks can be automated, but manual security testing is also necessary.

- **Tools Used:**

→ OWASP ZAP: Scans web applications for common security vulnerabilities.

→ Burp Suite: Helps test web application security.

→ SonarQube: Detects certain security weaknesses and code quality issues through static analysis.

→ Trivy: Scans container images, dependencies, and other supported software artifacts for vulnerabilities.

- **Industry Practices:**
  Companies integrate

Date: Day:

security scans into development pipelines, check third-party dependencies, and conduct manual penetration testing for important systems.

COMPARISON

Quality Process	Main Purpose	Automation Level
Functional Testing	Verify features and requirements	High for repeatable tests
Performance Testing	Measure speed and stability	High for scripted tests
Data Testing	Verify data accuracy and consistency	High for defined rules
Security Testing	Detect security weakness	Partial to high, depending on test

ROLE OF AUTOMATION

- Automation tools execute predefined tests, compare actual results, generate reports and alert teams when problems are detected.

Date: Day:

- Tools such as Jenkins and GitHub Actions can automatically run these tests whenever developers update the code.

- However, automation cannot completely replace human judgement.

→ Conclusions

- Functional, performance, data and security testing are essential processes in modern software development.

- Automation tools help organization test software faster, detect defects earlier and maintain consistent quality.

- The best approach combines automated testing with manual investigation and continuous monitoring to deliver reliable, efficient, accurate and secure software.

→ References

- Selenium – official Documentation

- Apache JMeter website
- dbt Documentation

- OWASP ZAP website
- Great Expectations Documentation